

ARTIFICIAL INTELLIGENCE

CO3 – ASSESSMENT TOOL 2

Multiple Choice Questions with Answers & Explanations

Course Name	Artificial Intelligence
Course Code	CSA1711
Questions	30
Total Marks	30

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Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
- b) An agent that uses a knowledge base and logical inference to decide actions
- c) An agent that only follows pre-defined scripts
- d) An agent that learns from rewards

Correct Answer: b) An agent that uses a knowledge base and logical inference to decide actions

Explanation: A logical agent uses stored knowledge and inference rules to derive conclusions and choose actions.

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$
- b) $\neg P \vee Q$
- c) $\neg Q \rightarrow \neg P$ only
- d) Both b and c

Correct Answer: d) Both b and c

Explanation: $P \rightarrow Q$ is equivalent to $\neg P \vee Q$ and also to its contrapositive $\neg Q \rightarrow \neg P$.

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
- b) A conjunction of disjunctions (clauses)
- c) A conjunction of literals only
- d) A sequence of implications

Correct Answer: b) A conjunction of disjunctions (clauses)

Explanation: CNF is an AND of clauses, where each clause is an OR of literals.

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass
- b) Every entity that is a student passes
- c) If something passes, it is a student
- d) There exists at least one student who passes

Correct Answer: b) Every entity that is a student passes

Explanation: The universal quantifier $\forall x$ means the rule applies to every entity x .

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

Correct Answer: b) Make two logical expressions syntactically identical

Explanation: Unification finds substitutions for variables so that two expressions match exactly.

Q6. In the Resolution algorithm, deriving the empty clause $\square$ indicates:

- a) The knowledge base is consistent
- b) The negated goal is satisfiable
- c) A contradiction — the original goal is proven
- d) More clauses need to be added

Correct Answer: c) A contradiction — the original goal is proven

Explanation: In resolution refutation, deriving $\square$ means the KB plus the negated goal is contradictory, so the goal follows.

Q7. Forward Chaining is best described as a _____ strategy.

- a) Goal-driven / top-down
- b) Data-driven / bottom-up
- c) Heuristic-based depth-first
- d) Probabilistic inference

Correct Answer: b) Data-driven / bottom-up

Explanation: Forward chaining starts with known facts and repeatedly applies rules to derive new facts.

Q8. Backward Chaining begins reasoning from:

- a) All known facts toward a conclusion
- b) The goal and works backward to verify supporting facts
- c) A random clause in the knowledge base
- d) The resolution refutation tree

Correct Answer: b) The goal and works backward to verify supporting facts

Explanation: Backward chaining starts with a goal and searches for rules and facts that can support it.

Q9. Which step is NOT part of converting a sentence to CNF?

- a) Eliminate biconditionals ($\leftrightarrow$)
- b) Move negations inward using De Morgan's laws
- c) Apply Skolemization to remove existential quantifiers
- d) Distribute AND over OR

Correct Answer: c) Apply Skolemization to remove existential quantifiers

Explanation: Skolemization is used in FOL conversion, but it is not a standard step for propositional CNF conversion.

Q10. Knowledge Engineering in FOL involves:

- a) Training a neural network on domain-specific data
- b) Defining domain concepts, relations, and axioms formally in FOL
- c) Writing heuristic search strategies
- d) Encoding reward functions for reinforcement learning

Correct Answer: b) Defining domain concepts, relations, and axioms formally in FOL

Explanation: Knowledge engineering builds a formal representation of domain knowledge using concepts, relations, and rules.

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

- a) $\neg Q$ is true
- b) Q must be true
- c) P is uncertain
- d) The KB is inconsistent

Correct Answer: b) Q must be true

Explanation: Modus Ponens directly derives Q from $P \rightarrow Q$ and the fact P .

Q12. The Most General Unifier (MGU) is the substitution that:

- a) Is the most specific possible match
- b) Makes two terms identical with minimal variable binding
- c) Eliminates all variables from a clause
- d) Applies resolution to produce CNF

Correct Answer: b) Makes two terms identical with minimal variable binding

Explanation: The MGU makes expressions identical while imposing the least specific necessary substitutions.

Q13. Which inference method is more appropriate when a specific goal is known?

- a) Forward Chaining
- b) Truth Table method
- c) Backward Chaining
- d) A* search

Correct Answer: c) Backward Chaining

Explanation: Backward chaining is goal-driven, so it is suitable when a specific conclusion must be verified.

Q14. Propositional Logic differs from First Order Logic because it:

- a) Supports probabilistic reasoning
- b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions
- c) Is always more expressive
- d) Handles uncertainty better

Correct Answer: b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions

Explanation: Propositional logic represents propositions as true or false, while FOL can represent objects, relations, and quantifiers.

Q15. Resolution Refutation proves a goal G by:

- a) Directly deriving G from the KB using forward chaining
- b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$)
- c) Checking G in the truth table
- d) Applying Modus Tollens repeatedly

Correct Answer: b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$)

Explanation: Resolution refutation negates the goal and derives the empty clause to show that the negation is impossible.

Q16. Learning from observation in AI refers to:

- a) An agent receiving numerical rewards from the environment
- b) Improving performance by examining examples or experience from the environment
- c) Acquiring knowledge by following explicit symbolic rules
- d) Memorizing a fixed dataset without generalization

Correct Answer: b) Improving performance by examining examples or experience from the environment

Explanation: Learning from observation means improving behavior or knowledge from examples and environmental experience.

Q17. Inductive learning is best described as:

- a) Deriving specific conclusions from general rules
- b) Generalizing a hypothesis from specific training examples
- c) Memorizing training data exactly
- d) Applying search algorithms to find optimal policies

Correct Answer: b) Generalizing a hypothesis from specific training examples

Explanation: Inductive learning moves from specific examples to a generalized hypothesis.

Q18. In a Decision Tree, what does each internal node represent?

- a) A class label / output decision
- b) A probability distribution over outcomes
- c) A test on an attribute / feature
- d) A training data sample

Correct Answer: c) A test on an attribute / feature

Explanation: Each internal node tests an attribute or feature and branches according to its possible values.

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

- a) Requires large amounts of training data to generalize
- b) Learns only from numeric data using regression
- c) Uses prior domain knowledge to explain and generalize a single example
- d) Applies unsupervised clustering to find hidden patterns

Correct Answer: c) Uses prior domain knowledge to explain and generalize a single example

Explanation: EBL uses existing domain knowledge to explain why an example works and generalizes that explanation.

Q20. In statistical learning, a Naive Bayes classifier assumes:

- a) All features are conditionally dependent on each other given the class
- b) The data must follow a uniform distribution
- c) All features are conditionally independent given the class label
- d) Labels are determined by a reward function

Correct Answer: c) All features are conditionally independent given the class label

Explanation: Naive Bayes assumes conditional independence among features when the class is known.

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

- a) Using the EM algorithm to estimate missing values
- b) Training a deep neural network with dropout regularization
- c) Directly computing maximum likelihood estimates from observed data
- d) Applying reinforcement signals to update weights

Correct Answer: c) Directly computing maximum likelihood estimates from observed data

Explanation: With complete observed data, parameters can be estimated directly using maximum likelihood.

Q22. The Expectation-Maximization (EM) algorithm is used when:

- a) Training data is fully labeled and no variables are hidden
- b) An agent explores an unknown environment using trial and error
- c) The dataset contains hidden (latent) variables that cannot be directly observed
- d) Backpropagation fails to converge in a neural network

Correct Answer: c) The dataset contains hidden (latent) variables that cannot be directly observed

Explanation: EM alternates between estimating hidden variables and updating model parameters.

Q23. In a Neural Network, the activation function serves to:

- a) Initialize the weights of each neuron to zero
- b) Normalize the input dataset before training
- c) Introduce non-linearity to enable learning of complex patterns
- d) Store previously learned examples in memory

Correct Answer: c) Introduce non-linearity to enable learning of complex patterns

Explanation: Activation functions add non-linearity, allowing neural networks to model complex relationships.

Q24. Which of the following correctly describes backpropagation in a neural network?

- a) Weights are updated in the forward pass using input gradients
- b) Outputs of each layer are stored and reused in later epochs
- c) The error is propagated backward from output to input layers to adjust weights
- d) A separate model is trained for each hidden layer independently

Correct Answer: c) The error is propagated backward from output to input layers to adjust weights

Explanation: Backpropagation calculates error gradients backward through the network and uses them to update weights.

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

- a) Minimizing the number of training examples used
- b) Grouping similar data points into clusters using centroids
- c) Finding the maximum-margin hyperplane that separates classes
- d) Updating weights using Q-values from an environment

Correct Answer: c) Finding the maximum-margin hyperplane that separates classes

Explanation: SVM seeks a decision boundary with the maximum margin between classes.

Q26. The kernel trick in kernel machines (SVMs) allows:

- a) Reducing the number of support vectors needed
- b) Accelerating gradient descent convergence
- c) Operating in a high-dimensional feature space without explicitly computing coordinates
- d) Replacing hidden layers in a deep neural network

Correct Answer: c) Operating in a high-dimensional feature space without explicitly computing coordinates

Explanation: The kernel trick computes similarity in an implicit high-dimensional feature space without explicitly mapping each point.

Q27. In Reinforcement Learning, the agent aims to maximize:

- a) Classification accuracy on a fixed test set
- b) The number of features selected from the training data
- c) Cumulative reward received over time through interaction with the environment
- d) The size of the neural network hidden layer

Correct Answer: c) Cumulative reward received over time through interaction with the environment

Explanation: The objective of reinforcement learning is to maximize long-term cumulative reward.

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

- a) A supervised learning algorithm requiring labeled datasets
- b) An unsupervised clustering technique using centroids
- c) A model-free, off-policy temporal difference learning method
- d) A decision-tree-based ensemble classifier

Correct Answer: c) A model-free, off-policy temporal difference learning method

Explanation: Q-learning learns action values directly from interaction without requiring an explicit environment model and follows an off-policy TD approach.

Q29. Overfitting in machine learning occurs when:

- a) The model performs poorly on both training and test data
- b) The learning rate is set too low during gradient descent
- c) The model fits training data too closely and generalizes poorly to new data
- d) The training set has more features than training examples

Correct Answer: c) The model fits training data too closely and generalizes poorly to new data

Explanation: An overfit model memorizes training patterns/noise and performs poorly on unseen data.

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

- a) RL uses labeled input-output pairs; SL uses reward signals
- b) Both RL and SL require explicit reward functions
- c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples
- d) RL is always faster to converge than SL on any dataset

Correct Answer: c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples

Explanation: RL learns through interaction and rewards, whereas supervised learning learns from labeled input-output examples.